Berkeley and Heiserman as an Unexhausted Architecture for Embodied Machine Intelligence

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Abstract

Edmund C. Berkeley is usually remembered as a writer who helped connect symbolic logic to computing machinery. That description is correct, but incomplete. Read across Berkeley's machine-oriented writings and projects, the central concern is broader: to show that symbolic logic can serve as a practical design language for machines that acquire information, retain it, respond appropriately to changing conditions, and organize their behavior over time. This paper takes *Symbolic Logic and Intelligent Machines* as the principal text in that effort, while treating it as representative of a larger Berkeley program that links logical form, circuitry, control, and intelligent behavior. On this reading, Berkeley does not merely treat intelligence as abstract symbol manipulation. He repeatedly defines intelligent machines in operational terms, ties logic to hardware realization, and describes machine behavior through the coordinated interaction of inputs, outputs, memory, calculation, control, states, and events. His treatment of robots and machine activities accordingly moves beyond static logical form toward temporally extended, environment-coupled behavior. David L. Heiserman's machine-intelligence work extends this program toward adaptive creature architectures built around memory, confidence, and generalization. Taken together, Berkeley and Heiserman can be read not as exhausted historical episodes but as contributors to a still under-tested architectural approach to embodied robotic cognition.

1 Introduction

Contemporary AI and robotics research often oscillates between large-scale statistical learning, abstract planning systems, and highly specialized control solutions. In that environment, behavior-organizing approaches rooted in explicit states, logical conditions, hardware-near control, and modular sensorimotor structure can be dismissed too quickly as dated. This paper begins from the opposite suspicion: that these design commitments remain fresh, technically coherent, and experimentally available for robotic cognition. The question is not whether twentieth-century machine theories should be repeated unchanged, but whether certain older design commitments still describe a usable architecture for embodied intelligence. Current discussion often overlooks that architecture because it is distributed across logic, control, robotics, and early machine-intelligence writing rather than consolidated under a single contemporary label.

Edmund C. Berkeley is commonly placed in the history of symbolic logic, early computing, and machine thought. That placement is justified. *Symbolic Logic and Intelligent Machines* is overtly concerned with Boolean algebra, logical relations, class reasoning, and the design of information-handling machinery [1]. Yet this standard placement risks narrowing Berkeley's significance. He was not only interested in the formal calculability of statements and classes. He was also interested in how reason-like competence could be built into machinery that senses, stores, decides, and acts. That broader concern is easy to miss if Berkeley is read only through the lens of formal logic. It becomes clearer when the same book is read as a document about machine organization: what kinds of parts an intelligent machine has, how those parts relate, how control is represented, and how behavior unfolds through states and events.

The aim of this paper is to argue that Berkeley should be read as an architect of embodied machine intelligence rather than only as a symbolic logician of machines. He is not a full theorist of adaptive AI in the Heiserman sense, but neither is he adequately understood as a purely formal or disembodied thinker. His work gives a logically rigorous account of how physically organized machines can take in information, retain it, operate on it, and guide their own activity over time [1, pp. 66–70]. When read alongside Heiserman's creature-based machine intelligence, Berkeley becomes part of a still-open design program rather than a closed episode in the history of ideas. The paper therefore asks the reader to shift levels of attention: away from Berkeley as only a contributor to symbolic notation and toward Berkeley as a thinker of machine architecture, behavior, and temporally extended control.

2 Research Question and Thesis

The guiding question is how Berkeley's *Symbolic Logic and Intelligent Machines* should be read today: merely as a period work of symbolic logic applied to machinery, or as a still-usable account of embodied, behavior-organizing intelligence. More analytically, the question is whether Berkeley's treatment of logic and machinery stays at the level of formal description, or whether it already specifies a machine architecture in which sensing, state, control, and action are treated as coordinated components of intelligence.

The working thesis is that Berkeley's work is best read as a technically viable account of embodied machine intelligence whose implications have not been exhausted. He inherits formal machinery from Boole, Venn, and Shannon, yet uses that inheritance to theorize intelligent systems as physically realized, temporally organized, and behaviorally active machines [2–5]. Heiserman extends that line toward adaptive creature intelligence organized around memory, confidence, and generalization [6]. More strongly, Berkeley and Heiserman together can be read as defining an underused alternative whose engineering potential has not been fully tested against current assumptions about how robotic intelligence must be built. For the purposes of this paper, that alternative can be stated schematically as a pipeline of sensing, state or memory, control, action, and adaptation. The literature review below is organized to support that stronger claim by showing that Berkeley is best understood neither as a routine symbolic logician nor as an isolated precursor, but as a figure whose work becomes sharper when placed among adjacent traditions in formal logic, cybernetics, adaptive machinery, and embodied control.

3 Literature Review

The literature relevant to this paper can be organized into five contributing bodies. The first establishes the logical and formal inheritance running from Boole and Venn through Berkeley and into the comparison with Heiserman. The second clarifies the mid-century symbolic-logic context against which Berkeley's distinctiveness becomes visible. The third places Berkeley within the broader cybernetic and machine-intelligence conversation shaped by Wiener, Ashby, Turing, and Walter. The fourth brings that discussion into contact with later embodied and behavior-based robotics. The fifth consists of secondary historical and philosophical interpretation, used here in a limited supporting role. Read in that order, the review moves from logical foundations, to contextual differentiation, to cybernetic placement, to embodied-robotic comparison, and finally to supporting interpretation.

The first body consists of the primary sources that directly carry the argument: Berkeley, Boole, Venn, Shannon, and Heiserman. Berkeley is the principal object of interpretation, but the point of the paper is not recoverable from Berkeley alone. Boole provides the underlying claim that logical operations can be written in a compact symbolic calculus and manipulated with rule-governed precision [2, 3]. Berkeley does not invent his symbolic resources from scratch; he inherits a tradition in which reasoning is already being recast as explicit operation. Boole gives him the confidence that classes, statements, and operations can be made exact enough to serve as components in a design language. Venn matters in two distinct ways. In *Symbolic Logic*, he sharpens the class-logic tradition Berkeley works within and clarifies the status of logical symbolism as a general calculational language rather than a mere classroom mnemonic [4]. Berkeley's use of symbolic logic as a practical language for machine organization sits more comfortably once that broader Vennian conception is in view. In *The Logic of Chance* and *The Principles of Empirical or Inductive Logic*, meanwhile, Venn moves toward probability, induction, and knowledge under conditions that are not exhausted by strict deduction [7, 8]. That second Vennian strand helps distinguish Berkeley's mostly rule-bound machine logic from Heiserman's move toward machines that store experience, accumulate success conditions, and generalize from them. Shannon supplies the crucial bridge from logical relations to relay and switching hardware: once logical form is realizable as circuitry, symbolic structure becomes directly architectural rather than merely notational [5]. If Boole and Venn explain Berkeley's formal inheritance, Shannon explains why Berkeley could treat logic as a machine-building discipline rather than only as a notational one. Heiserman, by contrast, is not part of Berkeley's own bibliographic world, but functions as a comparison point that clarifies how the same broad design space can be pushed from explicit control organization toward adaptive and experiential machine intelligence [6]. His Alpha, Beta, and Gamma levels supply a vocabulary for comparing fixed behavioral organization, learned response, and confidence-bearing generalization without abandoning embodiment. Heiserman therefore does not merely extend Berkeley chronologically; he pressure-tests Berkeley conceptually by asking what happens when a state-organized creature must also remember, prefer, and generalize. Taken together, these five primary figures supply the paper's minimal conceptual arc: formal logic, practical symbolism, hardware realization, machine organization, and adaptive extension.

The second body of literature consists of immediate contextual works in symbolic logic, especially Basson and O'Connor's *Introduction to Symbolic Logic* [9]. This source is worth more than a passing mention because it gives a fairly orthodox mid-century picture of what symbolic logic education looked like when centered on the propositional calculus, the algebra of classes, quantificational form, and formal deduction as such. Read against Basson and O'Connor, Berkeley's difference becomes sharper. He is not simply teaching symbolic manipulation, validity, or proof technique. He is repeatedly trying to move symbolic logic outward into circuit design, temporal organization, state transition, memory, and machine behavior. Without a foil like Basson and O'Connor, Berkeley's practical and architectural language can be mistaken for a routine extension of symbolic-logic pedagogy. With it, we can see that Berkeley is selecting from the symbolic tradition and reorienting it toward implementation, control, and intelligent machinery. Basson and O'Connor therefore help establish that Berkeley is unusual not because he knows different logical fundamentals, but because he asks those fundamentals to do a different kind of work. This contextual contrast is important to the paper's method: it lets the reader distinguish between Berkeley's inherited logical tools and Berkeley's distinctive use of those tools.

The third body of literature is the cybernetic and control literature: Wiener, Ashby, Turing, and Walter [10–15]. It prevents the present argument from sounding idiosyncratic or anachronistic. If Berkeley is read only against formal logic, his concern with control, behavior, and machine organization can appear eccentric or marginal. Read against cybernetics, those same concerns become legible as part of a broader technical conversation about regulation, signaling, adaptation, and machine activity in embodied systems.

Each author contributes something different to that clarification. Wiener contributes the most general postwar frame of communication, control, feedback, and the parallel treatment of animal and machine systems. He helps define the field in which organized signaling and regulation become central engineering questions. In relation to the present paper, Wiener is less a direct source for Berkeley than a way of situating Berkeley's machine concerns within a wider problem space where behavior and control are already central.

Ashby's two books are particularly important for this paper's vocabulary. *Design for a Brain* is directly relevant because it treats adaptation, equilibrium-seeking, and self-organizing behavior as machine problems rather than as mysteries outside mechanism [11]. *An Introduction to Cybernetics* adds a more explicit language of state, transition, regulation, variety, and machine description [12], making it especially useful for comparison with Berkeley's own state-and-event formalism. Ashby is especially important because he and Berkeley share an interest in state-organized systems while differing over emphasis: Ashby tends to abstract toward general regulatory principles, whereas Berkeley keeps pulling the discussion back toward logical specification, circuit organization, and the engineered composition of intelligent machines. The contrast helps locate Berkeley more precisely. He is not simply another cybernetic theorist of regulation, but one whose preferred idiom remains logical, design-oriented, and close to implementation.

Turing enters this literature review for a different reason. In *Intelligent Machinery* and "Computing Machinery and Intelligence," he keeps open the problem of machine intelligence as something that may require organization, education, learning, and developmental method rather than mere fixed calculation [13, 14]. His language of discrete machinery, memory capacity, and even unorganized machines gives us a second comparison class in which machine intelligence is treated as buildable, but not primarily through Berkeley's route of symbolic control architecture. Turing is therefore close enough to Berkeley to be illuminating and different enough to be clarifying: Berkeley is more architectonic and control-explicit, Turing more open to training, programming, and developmental organization. That difference shows that early machine-intelligence thought already contained divergent architectural bets. Berkeley's line is not the only one available, but it is one of the clearest for readers interested in explicit structure.

Walter's role is more concrete and embodied. His work on brain mechanisms and autonomous tortoise-like devices shows how relatively simple circuitry can produce environment-facing, apparently purposive behavior [15]. Walter matters because he demonstrates in physical form what Berkeley often argues in conceptual and architectural terms: that intelligence can appear at the level of organized sensorimotor behavior without requiring disembodied rationality. If Wiener provides a field, Ashby a state vocabulary, and Turing a contrasting developmental orientation, Walter provides a physically memorable demonstration of why embodied control should be taken seriously at all.

Taken together, these works do not prove Berkeley's argument, but they do show that Berkeley belongs in a live postwar conversation about regulation, autonomy, adaptation, and machine behavior, not only in the narrower history of formal logic. More importantly, they help the reader separate three questions that are often run together: whether intelligence can be mechanized, whether it must be embodied, and whether its organization should be explicit or learned. Berkeley matters in this landscape because he offers a strong answer to the first two questions and a distinctive answer to the third.

The fourth body of literature is behavior-based or embodied-machine comparison, especially Braitenberg and Brooks [16–18]. It gives the reader a later and more familiar contrast class. Many contemporary readers approach embodied intelligence through behavior-based robotics, situated action, or anti-representational traditions. Without addressing that lineage, Berkeley's embodied features may either be overlooked or misdescribed.

These references are used cautiously, but they should not be treated lightly. Braitenberg is valuable because he shows how surprisingly rich behavior can arise from very simple sensorimotor couplings. This keeps the paper from assuming that intelligence must begin at the level of high symbolic representation. Braitenberg's value here is diagnostic: he reminds us that embodied intelligence can emerge from control organization that is behaviorally effective even when it is structurally simple.

Brooks matters because he explicitly argues for situated, layered, behavior-producing architectures that do not depend on heavy centralized representation. The point is not that Berkeley anticipates Brooks in a straightforward way; he does not. Berkeley is much more willing than either Braitenberg or Brooks to preserve an explicit language of logical condition, internal organization, and designed control. Yet the comparison is still productive because it reveals that Berkeley's machines are not static theorem engines. They are systems that sense, transition, remember, and act in time. Brooks thus helps sharpen a negative and a positive point at once: negatively, Berkeley is not simply a forerunner of later behavior-based robotics; positively, he is also not confined to a disembodied symbol-processing picture of intelligence.

The later behavior-based literature therefore does two things at once: it prevents us from mistaking Berkeley for a contemporary behavior-based roboticist, and it prevents us from misclassifying him as a purely disembodied symbolicist. The resulting picture is more exact. Berkeley occupies a middle position in which symbolic organization and embodied control are not adversaries but cooperating layers of machine intelligence. That positioning matters because the paper's claim depends on precisely this middle ground. If the reader misses that point, the Berkeley–Heiserman comparison can collapse in either direction: Berkeley becomes too formal to matter for robotics, or too loosely embodied to matter for logical architecture. This fourth body of literature helps keep both distortions in check.

The fifth body of literature consists of secondary historical and philosophical interpretation. That body remains valuable, not marginal, because historical reconstruction and philosophical clarification are among the very means by which under-registered lines of thought become visible again. This paper contributes to that methodological point as well as to the substantive Berkeley–Heiserman argument. One reason the connection developed here has been easy to miss is that it falls between disciplinary narratives that are usually kept apart. By reconstructing the line across logic, cybernetics, machine intelligence, and contemporary embodied robotics, the paper is not only making a claim about Berkeley and Heiserman. It is also demonstrating a way of analyzing partially hidden continuities in the history of ideas and engineering practice. The present argument therefore does not reject further historical or philosophical work. On the contrary, it invites more of it. What it claims is that primary-source analytic reconstruction can itself be a productive method for recovering lines of thought that standard classifications leave underdescribed.

This review also clarifies the paper's novelty claim. In the literature assembled here, Berkeley and Heiserman usually appear in separate conversations: Berkeley in symbolic logic and machine thought, Heiserman in adaptive robot intelligence, and recent state-based robotics in a largely contemporary engineering register. On that basis, the present argument is not offered as a summary of an established Berkeley–Heiserman literature, but as a synthetic claim that emerges by reading these bodies of work together. The point of assembling these five bodies of literature is therefore not just to provide background. It is to show why that linkage is interpretable, technically serious, and worth stating as a live architectural claim.

From Interpretation to Engineering Claim

The argument of this paper is not merely that Berkeley and Heiserman deserve renewed historical attention. The stronger claim concerns design. Berkeley describes intelligent machinery in terms of logical conditions embodied in hardware, organized control, stored information, state transition, and purposive behavior. Heiserman extends that field toward machine creatures that retain successful responses and later generalize from them under confidence-sensitive conditions [6, pp. 18–20]. Read together, they define a coherent engineering line in which intelligence is built from organized relations among sensing, internal state, control, memory, and outward action. The remainder of the paper treats Berkeley and Heiserman against the common schema introduced above: sensing, state or memory, control, action, and adaptation.

What makes this line interesting now is not antiquarian recovery but systematic coherence. Both writers assume that intelligent machinery must be organized around explicit states, condition-sensitive transitions, sensorimotor coupling, and the coordination of internal and outward behavior. In Berkeley, these elements appear as logical conditions embodied in circuitry, control units, stored information, and behavior programs. In Heiserman, the same broad elements are retained but pushed toward creatures that modify later action in light of remembered success. Even where the two differ, they differ along a productive axis rather than across a conceptual break. Berkeley provides more formal and engineered control organization, while Heiserman pushes further toward adaptive behavior and memory-driven response. Read together, they make visible a design space that contemporary AI often treats separately under symbolic reasoning, control, behavior, and learning.

This does not yet prove that their architecture offers the best path forward for modern robotics. It does, however, justify treating it as a live engineering hypothesis rather than a completed or obsolete episode. The real question is not whether their systems can simply be reproduced, but whether state-based, hardware-near forms of behavior organization remain powerful in domains where deterministic timing, embodied interaction, modular design, and constrained compute still matter. If they do, then Berkeley and Heiserman become more than historically interesting.

Logic Embodied in Machinery

One of Berkeley's major contributions is to insist that symbolic logic is not merely a way of describing reasoning in abstraction. It is also a way of designing machinery. The bridge here is Shannon. Once Boolean relations can be realized in relay and switching circuits, conditions and inferences no longer remain on paper or in thought alone. They become structures of hardware [5]. In the schema used in this paper, this is the control layer stated in logical form and embodied in mechanism.

Berkeley adopts this possibility as fundamental. In his treatment, classes, statements, and logical conditions become design elements for circuits, control systems, and automatic machines. Logic is therefore not simply explanatory; it is operational. It specifies distinctions a machine can embody, conditions it can test, and sequences it can regulate. Berkeley's interest in symbolic logic is inseparable from his interest in machinery capable of carrying out organized, reason-like operations in the world [1]. At this point in the argument, Berkeley is supplying the formal basis on which sensing inputs, stored states, and action rules can be coordinated within a single machine design. This is one reason the paper insists on reading Berkeley architecturally. Without that shift in emphasis, the transition from logical relation to organized machine behavior can look like an optional application rather than a central feature of the book's project.

Intelligent Machines as Organized Systems

The strongest evidence for an embodied reading appears in Berkeley's own definitions and examples. He lists a wide range of machines as "intelligent machines," including traffic-light controllers, air-traffic systems, factory control equipment, game-playing machines, reading and recognizing devices, robots, and digital computers [1, pp. 66–69]. This list defines the target class broadly. Intelligent machinery is not confined to formal problem solving. It includes systems coupled to signals, environments, movement, and regulated action.

Berkeley then proposes a general architecture shared by almost every intelligent machine. Such a machine, he says, has inputs, outputs, storage or memory, calculating capacity, and a control unit [1, p. 70]. In the terms of this paper, that architecture already contains the major elements of the proposed schema: sensing through inputs, state or memory through storage, control through the control unit, and action through outputs. Especially important is Berkeley's remark that the control unit may itself be subject to instructions calculated by the machine, allowing the machine in some sense to guide itself [1, p. 70]. The autonomy is limited, but it means that control is not merely imposed from outside; it can be organized internally as part of the machine's own operation. This passage deserves emphasis because it prevents a common misreading. Berkeley is not merely listing machine parts. He is specifying a coordination problem: how perception, memory, processing, and directive structure are assembled so that behavior can be produced and maintained.

States, Events, and Temporally Extended Behavior

If Berkeley had stopped with Boolean algebra and machine architecture, the case for an embodied reading would remain incomplete. It becomes stronger in his treatment of the algebra of states and events. There Berkeley explicitly argues that Boolean calculus must be extended in order to come "into closer relation with the real world," since classes and statements change with time and since machine operation involves changing states and events rather than static truths alone [1, pp. 145–147]. This is the point at which the schema acquires an explicitly temporal account of state transition.

A state lasts for a time; an event is brief. This distinction allows Berkeley to formalize persistence, interruption, transition, delay, and sequence. He also insists on the practical importance of time units, step-functions, and even intervals in which a machine state is temporarily undefined during transition [1, pp. 145–147]. Once machine activity is represented as passage through states and events, intelligent behavior becomes describable as organized temporal process rather than as a mere arrangement of timeless logical forms. In the present argument, this is the component that connects stored condition, control logic, and temporally ordered action.

Robots and Behavioral Programs

Berkeley's robot chapters make the behavioral implications even clearer. He defines a robot as a machine able to "behave" by itself, taking in sensations by means of sensing devices, performing actions by means of acting devices, and correlating sensations and actions by means of circuits or mechanisms expressing instructions [1, pp. 177–178]. This definition explicitly links sensing, control, and action. The robot is not presented as a proof engine with motors attached. It is presented as a behaving system whose intelligence lies in the organization of sensation, action, and instruction. That formulation matters because it shifts the object of analysis from isolated inference to organized conduct. Once that shift is made, Berkeley's robot examples cease to look like colorful supplements and start to look like direct evidence for the paper's architectural reading.

His examples are equally revealing. Berkeley's electronic robot squirrel "Squee" is organized around activities such as hunting, homing, and depositing, each of which persists until relevant conditions terminate it [1, pp. 177–180]. Likewise, the "robot animal" problem is introduced explicitly as a programming problem in which one must assign a number of states to a robot and then specify changes from one state to another when certain events happen [1, pp. 177, 180–181]. These are behavior-organizing schemes in which logical conditions select, maintain, and redirect activity. In the shared schema, they are explicit instances of sensing coupled to state, control, and action over time.

Berkeley and Heiserman

The comparison with David Heiserman clarifies both Berkeley's strengths and his limits. Heiserman's *Robot Intelligence ... with Experiments* presents an evolutionary hierarchy in which Alpha-class creatures rely on constrained reflexive activity, Beta-class creatures add a memory of past successful responses, and Gamma-class creatures generalize from prior experience while building confidence in successful patterns of action [6, pp. 18–20]. In that framework, machine intelligence is lived by a creature in an environment. It senses, acts, remembers, adapts, and in the Gamma case anticipates. This matters for the present paper because it lets us compare Berkeley and Heiserman at the level of architecture rather than reputation: both are concerned with machines whose behavior depends on structured relations among input, internal organization, stored information, and action, but Heiserman adds explicit mechanisms for adaptive revision. In the shared schema, adaptation is the element that Heiserman develops most explicitly. He thus supplies what Berkeley mostly leaves implicit: a developmental account of how embodied organization can become progressively more adaptive.

Berkeley does not reach this level of adaptivity. His machines are mostly engineered top-down rather than self-modifying through reinforcement or confidence-based generalization. He does not provide Beta-style stored response schemes or Gamma-style conjectural extensions of prior success.

Even so, the continuity remains important. Berkeley already supplies the control-and-behavior side of the story: intelligence as physically realized organization, behavior as transitions among activities, sensing and acting as core capacities, and internal control as the basis of outward competence. Heiserman does not replace that framework so much as deepen it by adding mechanisms for experiential revision. This is the hinge of the comparison. If Berkeley already gives us a machine architecture for embodied control, then Heiserman can be read not as a disconnected later enthusiasm but as a more adaptive development within the same broad design space.

The difference, then, is not that Berkeley lacks embodiment. It is that Berkeley's version is primarily engineered and prescribed, whereas Heiserman's becomes increasingly adaptive and experiential. Berkeley links logic and circuitry to organized machine behavior; Heiserman pushes the same design space toward creature-like adaptation. Their relation is therefore best understood as continuity at the architectural level with divergence at the developmental one.

4 A Contemporary Use-Case

The paper's main contemporary contribution is to show that the Berkeley–Heiserman architecture is not only interpretable in historical texts but also operationally visible in a modern analysis setting. A companion 2026 study by the present author, *A Behavioral State Vocabulary in Sony ERS-111 R-CODE*, studies Sony's preserved 'R-CODE' sample corpus for the 'ERS-111' AIBO and argues that many apparently distinct routines are organized by a compact recurring behavior vocabulary [19]. The function of that companion paper here is not to serve as independent external confirmation. It is to develop, in a constrained robotic corpus, a concrete analytic application of the same line of thought. Its importance for the present argument is that it provides a setting in which behavior organization can be analyzed, compared, and reused at the level of design.

In the terms developed earlier in this paper, the contribution of the 'R-CODE' analysis is to make the common schema empirically legible. Across the sample set, routines can be read not merely as flat command sequences but as structured combinations of initialization, sensing, state retention, decision, action, synchronization, recovery, and repetition [19]. That is already close to Berkeley's machine description. Inputs, outputs, storage, control, and behavior appear here in practical script form. It also opens a path toward Heiserman's side of the comparison, because once such structures are made explicit, one can ask which parts are fixed, which are mode-based, which are responsive to sensed conditions, and which could in principle be extended toward adaptive revision.

The strongest value of this use-case lies in the shift from single-script reading to corpus-level abstraction. The claim of the companion study is not simply that one or two AIBO routines can be redescribed after the fact as state machines. Rather, the argument is that many sample programs share a small embodied grammar centered on startup conditions, environmental or bodily sensing, conditional branching, iterative motor activity, synchronization, and recovery [19]. In practical terms, this means that behaviors such as locomotion demos, obstacle-avoidance routines, and ball-tracking scripts can be compared within one analytical frame instead of treated as isolated scripts. That move matters for the present paper because it supplies exactly the kind of intermediate level of description that Berkeley's state-and-event formalism calls for: not raw machine code, not disembodied symbolism, but organized behavioral structure.

The 'R-CODE' corpus is especially useful because it spans a graded range of behavioral complexity and now makes that grading explicit. At the simpler end are activation or playback-like routines that largely sequence prepared actions. More complex cases add persistent monitoring and bodily recovery, as in locomotion scripts that watch fall state and trigger restorative behavior. Still richer cases add local branching, environmental comparison, and eventual mode-governed task control, as in maze navigation and football behaviors [19]. This ladder of examples makes it possible to observe, within one preserved platform, a progression from startup regularization, to monitored action, to decision loops, to fuller state-governed behavioral organization. That progression is directly relevant to the present argument, because it shows how explicit states, sensory conditions, control branches, and recurrent actions combine into a usable design vocabulary for embodied intelligence.

Two brief examples make the point more tangible. A script such as 'Move' can be read at the behavioral level as: boot, assume a safe pose, repeat forward motion, monitor fall state, and invoke recovery if needed. In the extracted analysis, this routine contains four states and five transitions, and its sensed variable is the fall indicator 'Gsensor_status' [19]. In the paper's schema, that is already a nontrivial organization of action, sensing, control, and recovery rather than a bare motion sequence. The script does not simply command the robot to walk. It couples repeated action to bodily monitoring and to a distinct restorative path, which means that the behavior is organized around conditional continuity rather than around one uninterrupted motor command.

A richer case such as 'Football' makes the state structure more explicit. At the behavioral level, the routine can be summarized as search, detect ball, pursue, kick, and resume, with fall recovery and tracking loss handled as separate branches [19]. The local extract identifies thirty-one states, fifty-four transitions, sensed variables including 'Cdt_npixel', 'Gsensor_status', 'Head_pan', 'Head_tilt', and 'Psd_range', and explicit internal variables such as 'mode', 'head', and 'lost' [19]. What matters here is not the surface task alone but the mode structure underneath it. Search and pursuit are not just different motions. They are distinct behavioral states linked by sensor conditions, maintained by internal variables, and re-entered when the environment changes. This is precisely the level at which the paper's Berkeley–Heiserman claim becomes concrete: the routine is not well described as a bag of commands, but as an organized cycle in which sensing, temporary state, branching, action, and recovery cooperate over time.

The corpus becomes even more informative when several scripts are compared together. 'C-Tracking' is close to capability activation: a minimal startup-and-act pattern. 'Move' adds explicit monitoring and recovery. 'Maze' marks the important intermediate threshold at which the behavior becomes visibly decision-structured, because it repeatedly samples local conditions, compares alternatives, and reorients under environmental constraint. 'Football' extends that same logic into persistent mode structure and a fuller embodied state machine [19]. Presented in that order, the sample set yields more than isolated examples. It yields a compact comparison ladder in which one can watch the growth of behavioral organization from simple activation, to monitored action, to decision loops, to explicit mode-based control. That progression is itself part of the paper's contribution, because it makes the architecture visible as a graded design vocabulary rather than as a single all-or-nothing theory.

Script	States	Sensing	Recovery	Mode/Branching
‘C-Tracking‘	1	none explicit	none	minimal activation
‘Move‘	4	‘Gsensor_status‘	explicit	monitor loop
‘Maze‘	12	‘Distance‘, ‘Head_pan‘	retry/escape	decision loops
‘Football‘	31	vision, posture, range	explicit	‘mode‘, ‘head‘, ‘lost‘

Even in this compressed form, the comparison makes a concrete point that is easy to miss in script-by-script reading: the corpus is not just varied in theme, but ordered in control complexity. The progression from one state and no explicit transition structure in ‘C-Tracking‘, to monitored looping in ‘Move‘, to environmental comparison and controlled reorientation in ‘Maze‘, to multi-variable mode control in ‘Football‘, gives the reader a visible scale along which the behavior vocabulary grows.

That pedagogical point should be stated explicitly. For readers encountering this material for the first time, the jump from a compact monitored routine such as ‘Move‘ to a dense mode-driven routine such as ‘Football‘ can be too abrupt if the intermediate cases are not given enough space. The present paper therefore uses the ladder not only as evidence but also as a reading sequence. In that sequence, ‘Maze‘ does the crucial bridging work: it shows how monitored action becomes decision-structured behavior before the reader is asked to absorb the richer mode logic of ‘Football‘. A fuller companion treatment of the AIBO corpus should continue expanding the intermediate cases so that the method of analysis is learned as a progression of increasingly rich behavior blocks rather than inferred from a leap between simple and highly structured examples.

The point here is not to claim direct historical descent from Berkeley to Sony or from Heiserman to ‘R-CODE‘. Nor is it to treat one paper by the present author as independent confirmation of another. The evidential function of the section is different. It shows that the ingredients isolated in the earlier parts of the paper remain available as engineering abstractions in a contemporary analysis context. When the AIBO corpus is read through a behavioral state vocabulary, Berkeley's emphasis on temporally organized machine activity becomes legible again, and Heiserman's emphasis on creature-level organization becomes easier to operationalize as an extension problem rather than as a separate conceptual universe.

If a preserved constrained robotic corpus can be redescribed in terms of stable behavior blocks, recurring state transitions, recovery paths, and mode-based embodied control, then the Berkeley–Heiserman line is not merely of historical interest. It becomes a tractable research program. One can analyze existing behaviors, compare them across a corpus, extract a reusable control vocabulary, and then ask how far such architectures can be extended toward richer forms of adaptive embodied intelligence. On that basis, the contemporary use-case does more than illustrate relevance. It supplies a methodological bridge between historical interpretation and technical reconstruction [19].

5 Conclusion

Berkeley's contribution can be stated in technical terms. He treats intelligent machinery as an organized relation among inputs, outputs, storage, calculation, control, states, events, and behavioral programs. On that basis, Berkeley is more than a symbolic logician applying formal methods to machines. He provides a control-oriented account of how intelligent conduct can be specified, embodied, and temporally organized in machinery.

The comparison with Heiserman sharpens both the reach and the limit of that account. Berkeley does not formulate adaptive, reinforcement-based, or confidence-driven machine intelligence in the Heiserman manner. What he does provide is the groundwork: sensing, internal state, control, and action organized in explicit machine form. Heiserman extends that framework by developing the adaptive component more fully through remembered success, generalization, and creature-level learning behavior.

The result is a tractable research program rather than a completed doctrine. Berkeley and Heiserman together define a framework in which logic, explicit state organization, embodied control, behavioral modularity, and adaptive response are treated as compatible components of machine intelligence. Contemporary work on behavioral state vocabularies for constrained robotic systems supports the claim that this framework remains technically relevant [19]. The next step is not further historical recovery but systematic experimental evaluation of this framework against contemporary assumptions about how embodied intelligent systems should be built.

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